

SUBJECT: CHEMISTRY-2019

Max. Marks-70

Time: 3.15 Hrs

INSTRUCTION

This question paper has 4 parts, all parts are compulsory.

- **Part-A carries 10 marks. Each question carries one mark.**
- **Part-B carries 10 marks. Each question carries one mark.**
- **Part-C carries 15 marks. Each question carries one mark.**
- **Part-D carries 15 marks. Each question carries one mark.**

PART-A

I. Answer all the questions. Each question carries one mark

10x1=10

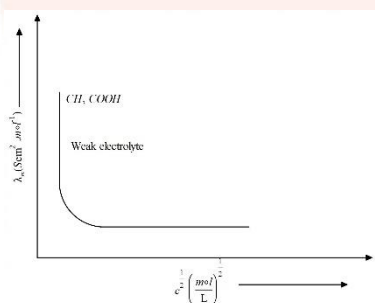
1. How does the size of blood cells change when placed in an aqueous solution containing more Than 0.9% (m/v) sodium chloride?

Sol. Shrink

2. How does the volume change on mixing two volatile liquids to form an ideal solution?

Sol. $\Delta V_{\text{mixing}} = 0$ (there is No change in the volume upon mixing the liquids)

3. Draw a graph of $\lambda_m \frac{v}{c} \sqrt{c}$ for acetic acid (weak electrolyte) solution.



Sol.

4. Unit of rate constant of a reaction is same as that of its rate. What is the order of the reaction?

Sol. Zero order reaction

5. Among physisorption or chemisorptions which one has higher enthalpy of adsorption?

Sol. Chemical adsorption or Chemisorption

6. What is the role of depressant (NaCN) in Froth-Flotation method?

Sol. In the case of an ore containing ZnS and PbS, the depressant used is NaCN. It selectively

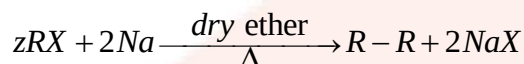
prevents Zns from coming to the froth but allows Pbs to come with the forth.

7. Name the noble gas having ns^2np^6 electronic configuration but does not have d-orbitals in its valence shell.

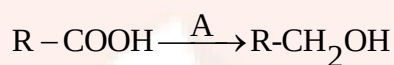
Sol. Neon

8. Write the general equation for Wurtz reaction.

Sol. Alkyl halide on heating with metallic sodium in dry ether gives higher alkane. This reaction is called Wurtz reaction.



9. What is the reagent 'A' used in the following equation?



Sol. $LiAlH_4$

10. Which vitamin deficiency causes the disease pernicious anaemia?

Sol. B_{12}

PART-B

II. Answer any five of the following questions.

5x2=10

11. Lithium metal has a body centred cubic lattice structure with edge length of edge unit cell 352 pm. Calculate the density of lithium metal. [Given: Atomic mass of Li = 7 $gmol^{-1}$ $N_A = 6.022 \times 10^{23} \text{ Atoms } mol^{-1}$]

$$\text{Sol. } d = \frac{2 \times 7}{(6.022 \times 10^{23})(352 \times 10^{-10})^3}$$

$$d = 0.53 \text{ g/ml}$$

12. State Faraday's Second law of electrolysis.

Sol. When same quantity of electricity passes through solutions of different electrolytes, the amounts of substances liberated at the electrodes are directly proportional to their chemical equivalent masses.

13. What is pseudo first order reaction? Give an example.

Sol. A reaction which appears to be of higher order but made to follow first order kinetics by taking the other reactants except one in large excess.



$$r \propto [CH_3COOC_2H_5][H_2O]^0 ; \text{order} = 1$$

14. How will you account for the following?

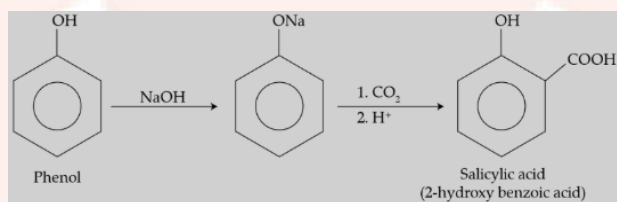
- i) Actinoids exhibit more number of oxidation states than lanthanoids.
- ii) Atomic radii of second and third transition series elements are almost identical.

Sol. i) As in actinoids the outer electrons are less firmly held, they are easily available for bonding.

ii) Due to lanthanoid contraction.

15. Explain the Kolbe's reaction with equation.

Sol.

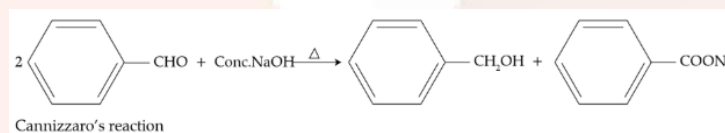


When phenol is treated with sodium hydroxide, sodium phenoxide is produced. This sodium phenoxide when treated with carbon dioxide, followed by acidification, undergoes electrophilic substitution to give ortho- hydroxybenzoic acid as the main product. This reaction is known as Kolbe's reaction.

16. Write the equation for the reaction between Benz aldehyde and concentrated NaOH

Solution. Name the reaction.

Sol.



17. i) What are anionic detergents?

ii) What is the role of saccharin in food?

Sol. i) In anionic detergents large part of the molecules is anion and it is the anionic

ii) Artificial sweetener.

18. Give one example each for the following.

i) Antifertility drugs.

ii) Narcotic analgesics.

- Sol.** i) Norethindrone, ethynylestradiol, novestrol
 ii) They produce sleep so they are used for the relief of post-operative pain, cardiac pain, labour pain in child birth etc. Example: Morphine, heroin, codeine etc.

PART-C

III. Answer any five of the following questions.

5x3=15

19. a) In the extraction of Aluminum by electrolysis,

(i) Write overall cell reaction.

(ii) What is role of cryolite?

Sol. a) (i) $2Al_2O_3 + 3C \rightarrow 4Al + 3CO_2$

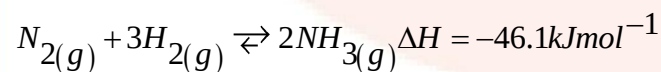
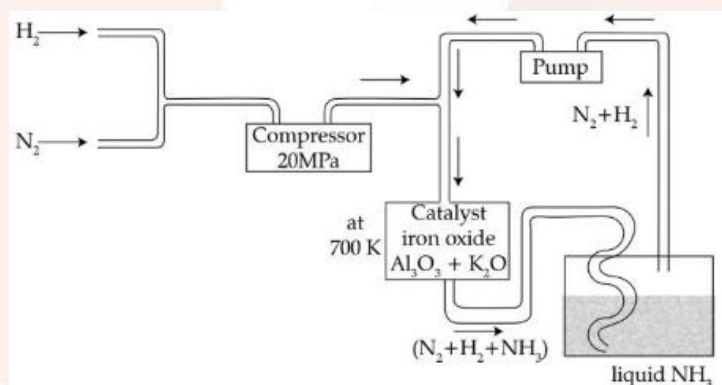
(ii) The molten cryolite decreases the melting point and also increases the electrical conductivity.

b) Name the metal refined by Mond's process.

Sol. Nickel

20. In the manufacture of ammonia by Haber's process. Write the flow chart and chemical equations with optimum conditions.

Sol.



The optimum condition of pressure of 200 bar and 750K are used along with the catalyst such as finely divided iron with small amount of molybdenum, or potassium oxide or aluminum oxide as promoter.

21. (i) Hydrogen bonding in H_2O but not in H_2S
(ii) $ConcH_2SO_4$ is a good dehydrating agent

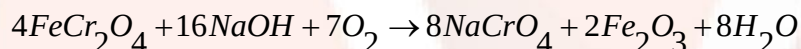
Sol. (i) In H_2O electronegative element like oxygen is present that involves I hydrogen bonding
(ii) $Conc H_2SO_4$ has high affinity for water.

22. Complete the following chemical equations:

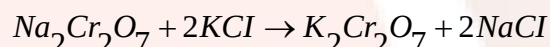
- (i) $NH_3 + 3Cl_2 \rightarrow \dots\dots + 3HCl$
(ii) $Na_2SO_3 + 2HCl \rightarrow 2NaCl + H_2O + \dots\dots$
(iii) $Br_2 + 3F_2 \rightarrow \dots\dots$

Sol. (i) NCl_3
(ii) SO_2
(iii) $2 Br F_3$

23. Write the balanced chemical equation involve in the manufacture of potassium-dichromate form chromite ore.



Sol. $2Na_2CrO_4 + conc.H_2SO_4 \rightarrow Na_2Cr_2O_7 + Na_2SO_4 + H_2O$



24. i) What are interstitial compounds?

ii) Transition metals show good catalytic property. Give any two reasons.

Sol. i) These are the compounds formed when the smaller atoms of H, C, N and B occupy the interstitial places in the metallic crystals of d-block.

ii) a) transition metals exhibit variable oxidation states due to the presence of unpaired d - electrons.

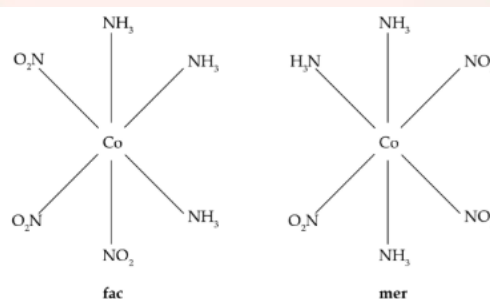
b) Finely powdered transition metals provide a large surface area.

25. a) Write the IUPAC name of $K_3[Cr(C_2O_4)_3]$.

b) Give the facial (fac) and meridional (mer) isomeric structures of $[Co(NH_3)_3(NO_2)_3]$.

Sol. a) Potassium trioxalatochromate (III)

(b)



26. With the help of valence Bond Theory (VBT) explain hybridization, geometry and magnetic property of $[\text{Ni}(\text{CN})_4]^{2-}$ tetracyanido nickelate (II) ion. [Given: Atomic number of Ni = 28].

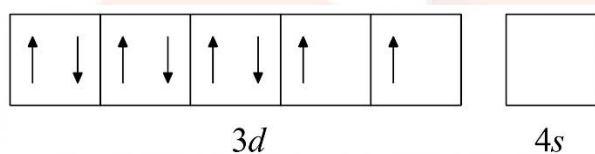
Sol. The electronic configuration of nickel ($Z = 28$) in the ground state is $[\text{Ar}] 3d^8 4s^2$. In the + 2 oxidation state, the configuration of nickel is $[\text{Ar}] 3d^8 4s^0$.

When cyanide ligands approach the Ni^{2+} ion the unpaired 3d electrons get paired up against Hund's rule leaving one of the d orbitals vacant.

$[\text{Ni}(\text{CN})_4]^{2-} : \text{Ni}(28) : 1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$ or, last shell EC is $3d^8 4s^2$

in $[\text{Ni}(\text{CN})_4]^{2-}$, Ni is in + 2 state

$\therefore \text{Ni}^{2+} : 3d^8 4s^0$



CN being a strong field ligand pairs up e^- s,

Thus

$[\text{Ni}(\text{CN})_4]^{2-} :$

Now dsp^2 hybridisation takes place giving four vacant hybrid orbitals directed to the four corners of a square. The electrons pairs donated by cyanide ligands occupy the dsp^2 hybrid orbitals of Ni^{2+} ion

Due to the absence of unpaired electrons, the inner orbital complex will be diamagnetic. The complex has a square planar shape.

PART-D

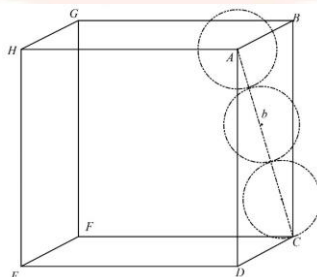
IV. Answer any three of the following questions.

3x5=15

27. a) Calculate the packing efficiency of in face centred cubic lattice

b) What is Frenkel defect? What is its effect on density of a solid?

Sol. a) In fcc : Let the edge length of unit cell = a

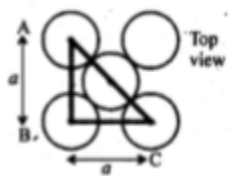


Let the radius of each sphere = r

$$\therefore AC = 4r$$

From right angled triangle ABC,

$$AC = \sqrt{AB^2 + BC^2} = \sqrt{a^2 + a^2} = \sqrt{2a^2} = \sqrt{2}a$$



$$\therefore \sqrt{2}a = 4r$$

$$\therefore a = \frac{4r}{\sqrt{2}}$$

$\therefore$ Volume of the unit cell

$$= a^3 = \left(\frac{4}{\sqrt{2}}r \right)^3 = \frac{64r^3}{2\sqrt{2}} = \frac{32r^3}{\sqrt{2}}$$

No. of unit cell in fcc = 4

$$\therefore \text{Volume of four spheres} = 4 \times \frac{4}{3}\pi r^3 = \frac{16}{3}\pi r^3$$

$\therefore$ Packing efficiency

$$= \frac{16\pi r^3/3}{64r^3/\sqrt{2}} = 0.74, \text{ i.e., } 74\%$$

b) Frenkel defect is the displacement of few smaller ions (generally cation is smaller than anions) from their normal position to the interstitial sites.

28. a) 31 g of an unknown molecular material is dissolved in 500 g of water. The resulting solution freezes at 271.14 K. Calculate the molar mass of the material. Given: K_f for water = 1.86 K K gmol^{-1} T_f° of water = 273 K.

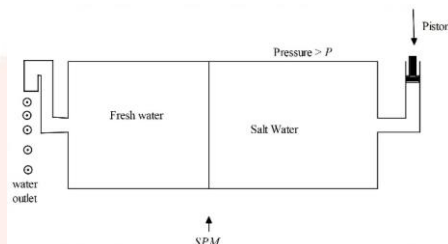
b) What is reverse osmosis? Mention its use.

Sol. a)
$$M_2 = \frac{K_f \times W_2 \times 1000}{\Delta T_f \times W_1}$$

$$= \frac{1.86 \times 31 \times 1000}{1.86 \times 500}$$

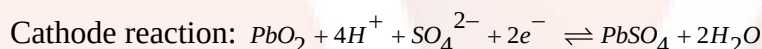
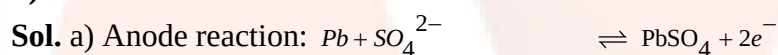
$$= 62 \text{ g/m}$$

b) The process of reversing the direction of osmosis by applying the pressure than the osmotic pressure to the solution of higher concentration is called reverse osmosis.



29. a) Write the equations for the reactions taking place at anode and Cathode in the lead storage battery.

b) Calculate the value of $\Delta_r G^\circ$ at 298K for the cell reaction.



b) $\Delta G^\circ = -nFE^\circ_{\text{Cell}}$

$$= -6 \times 96487 \times 0.7$$

$$= -405245.4 \text{ J}$$

$$= -405.2 \text{ kJ}$$

30. a) Derive integrated rate equation for a first order reaction

b) The specific reaction rate of a reaction water quadruples when temperature changes from 30°C to 50°C . Calculate the energy of activation of the reaction.

Sol. a) Consider $R \rightarrow P$

Integrating on both the side,

i.e., $\ln [R] = -kt + C \dots\dots\dots(1)$

C – constant of integration

To find the value of C

When $t = 0$, $[R] = [R]_0$

Substituting in (1)

$$\ln [R]_0 = -k(0) + C$$

$$C = \ln [R]_0$$

Substituting in equation 1

$$\ln [R] = -kt + \ln [R]_0 \dots\dots(2)$$

$$kt = \ln [R]_0 - \ln [R]$$

b)

$$\log \left(\frac{k_2}{k_1} \right) = \frac{E_a}{2.303R} \left[\frac{T_2}{T_1} - \frac{T_1}{T_2} \right]$$

$$\log 4 = \frac{E_a}{2.303} \times (323 - 303)$$

$$2.303 \times 8.314 \times 323 \times 303$$

$$E_a = 56.2 \text{ KJ/mol}$$

31. a) Define Shape Selective catalysis. Name the Zeolite Catalyst used to convert alcohols to gasoline in petroleum industry

b) What is peptisation? Give an Example

c) Write the expression for Freundlich adsorption isotherm

Sol. a) Zeolite are micro porous aluminosilicates with three dimensional network of silicates in which some silicon atoms are replaced by aluminum atoms giving Al-O-Si framework. ZSM-5

b) The process of conversion of freshly prepared precipitate into a colloidal solution by adding an electrolyte containing the common ion is called peptisation. Example Ferric hydroxide sol

$$\frac{x}{m} = kp^{1/n},$$

where x is the mass of the gas adsorbed on mass m of the adsorbent at pressure

P, k and n are constants which depend on the nature of the adsorbent and the gas at a particular temperature.

V. Answer any FOUR of the following questions.

4x5=20

32. a) Write the equation for the steps involved in the S_N1 mechanism of hydrolysis of 2-bromo 2- methyl propane.

b) i) Name the product formed for the reaction of isopropyl iodide on alcoholic KOH.

ii) What is the condition to be satisfied for a compound to be chiral?

c) What is racemic mixtures?

Sol. a) S_N¹ means unimolecular nucleophilic substitution reaction.

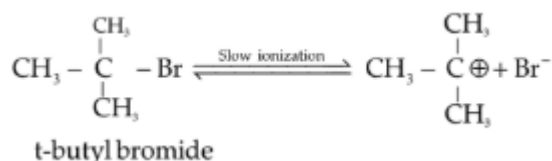
In this reaction rate depends only on the concentration of Alkyl halide. i.e. $ra[R-X]$

Where 'r' = rate of the reaction

$[R-X]$ = Molar conc. of alkyl halide

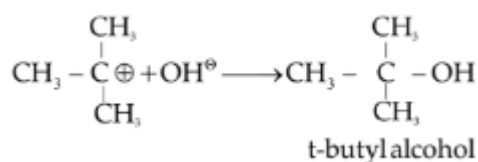
S_N¹ mechanism follows first order kinetics and the mechanism os of 2 steps.

Step 1:



Since, this step is slow step, it is rate determined step.

Step 2:



b) i) Propene

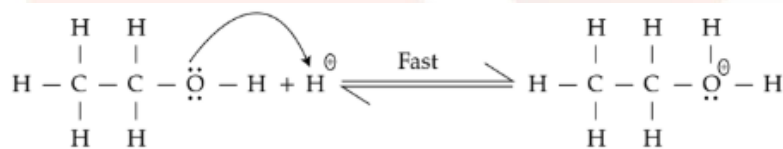
ii) The chiral molecule should contain asymmetric carbon atom.

c) Equimolar mixture of dextrotatory and levorotatory isomers is called racemic mixture.

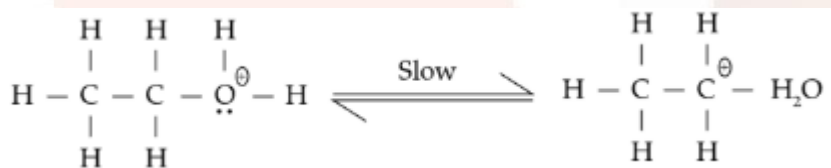
33. a) Explain the mechanism of acid catalyzed dehydration of ethanol to ethane.

b) How do you prepare methoxy ethane by Williamson's ether synthesis?

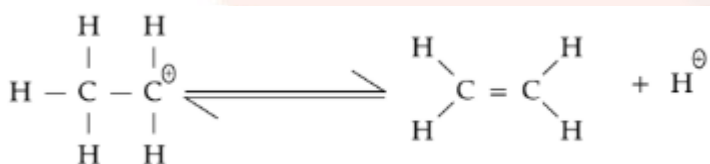
Sol. a) Step 1: Formation of protonated alcohol:



Step 2: Formation of carbocation:

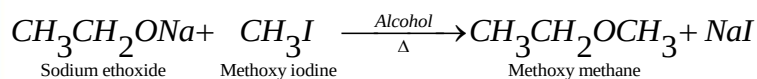


Step 3: Elimination of proton-



b) In a Williamson synthesis, an ether is formed by reacting an alcohol and an alkyl halide in the presence of a base.

When $\text{CH}_3\text{-I}$ is mixed with the base, like NaOH , the alcohol is deprotonated, leaving a negatively charged oxygen. This acts as a nucleophile and attacks the carbon bonded to the halogen (iodine). The halogen, a good leaving group, is released, leaving behind methoxy ethane.



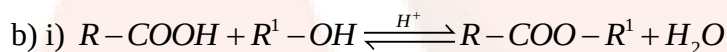
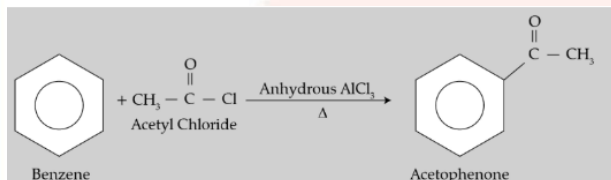
34. a) How does benzene reacts with acetyl-chloride in the presence of anhydrous AlCl_3 ? Give equation. (2)

b) i) Write general equation for esterification reaction.(1)

ii) Name the product obtained when benzoic acid is heated with ammonia.(1)

c) Name the reagent used in the Clemmensen reduction. (1)

Sol. a)



ii) Benzamide

c) Zn/Hg and conc. HCl

35. a) Between CH_3NH_2 and $\text{C}_6\text{H}_5\text{NH}_2$ which is more base? Give reason.

b) i) Name the main product when aniline is heated with alcoholic KOH and chloroform.

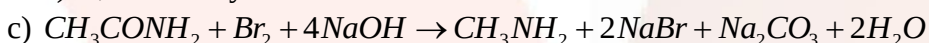
ii) Give the IUPAC name of $(\text{CH}_3)_2\text{N}-\text{C}_2\text{H}_5$

c) Complete the Chemical equation.

Sol. a) Methylamine is more basic because of electron releasing nature of methyl group

b) i) Phenyl isocyanide

ii) N,N-dimethyl ethanamine

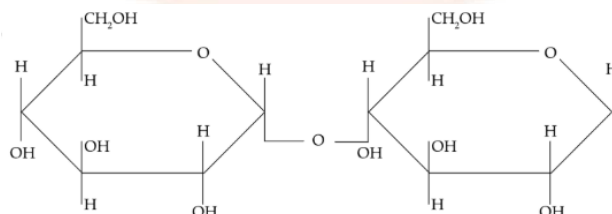


36. a) Write the Haworth structure of maltose.

b) What is peptide linkages? How peptide bonds are present in a tetra-peptide?

c) Name the hormone which regulates blood sugar level in the body.

Sol. a)



b) A peptide bond is formed by the reaction between amino groups of one amino acid molecule with carboxyl group of another amino acid molecule with the elimination of water. There are four peptide bonds in tetra peptide.

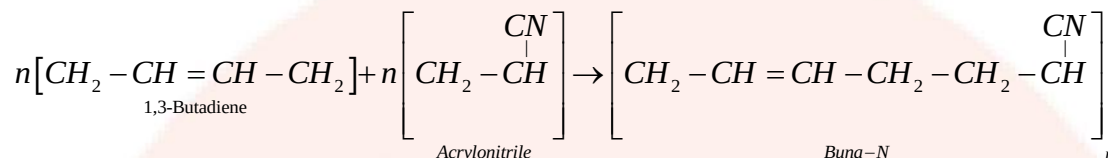
c) Insulin

37. a) How is Buna-N prepared? Give equation.

b) Name the monomers of Nylon-6,6.

c) What are thermosetting polymers?

Sol. . a)



b) Adipic acid and hexamethylenediamine

c) These are cross-linked or heavily branched polymers which got hardened during moulding process. These can't be softened again on heating.